Brief: Hallucination Detection in Graphs — Issues-FS Research

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from Human (project lead)
to Issues-FS team
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The Question

Can we flag hallucinations in a graph? Can we tag them, query them, visualise them, and measure them?

The short answer: yes, and Issues-FS is uniquely well-suited for this — better than a traditional property graph — because of its fractal structure and provenance-first design.

What Is a Hallucination (In Graph Terms)?

Strip away the LLM-specific connotations. At the most fundamental level:

A hallucination is a piece of information for which we cannot find evidence to support it.

When you follow the provenance — trace the edges back to their sources — you hit a dead end. There's no grounding. No evidence. No source. The information exists as an isolated node, or connects only to other ungrounded nodes.

In graph terms: **a hallucination is a node (or subgraph) with no valid provenance path to a trusted source.**

This applies to:

- LLM outputs (the obvious case)
- Human opinions stated as facts
- Data from unreliable sources
- Information that contradicts other established information
- Facts that were once true but are now outdated

The Spectrum Model

Hallucination is not binary. It's a spectrum:

```
GROUNDED                                    UNGROUNDED
   │                                             │
   ▼                                             ▼
[Verified fact]  [Evidence-backed]  [Opinion]  [Unfounded]  [Contradicted]  [Fabricated]
  with multiple    with some       stated     no          conflicts      made up,
  independent      supporting      clearly    evidence    with known     no basis
  sources          evidence        as such    found       facts          at all
```

At the left: a fact verified by multiple independent sources with clear provenance chains. At the right: a completely fabricated statement with no evidence and active contradictions.

Everything starts somewhere on this spectrum. The interesting work is in the middle — where evidence is partial, sources are ambiguous, and the confidence level is somewhere between "probably true" and "probably not."

In Issues-FS Terms

Every node in the graph can carry a **confidence/grounding score** based on:

Signal	What It Tells Us	Direction
Provenance depth	How many levels back can we trace this?	Deeper = more grounded
Source count	How many independent sources support this?	More = more grounded
Source credibility	How reliable are the supporting sources? (See Section 3)	Higher credibility = more grounded
Contradiction count	How many other nodes contradict this?	More contradictions = less grounded
Consistency	Does this fit with the surrounding graph?	Consistent = more grounded
Age / staleness	When was the evidence last verified?	Fresher = more grounded
Attribution	Is the source an LLM, a human, a system, a document?	Varies — but LLM without evidence = lower starting point

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The Provenance Dead End

The core detection mechanism. When you trace a piece of information backwards through the graph:

Healthy provenance chain:

```
Claim: "AES-256 is quantum-resistant at 128-bit effective strength"
  └── Source: NIST Post-Quantum Cryptography FAQ
      └── Source: Grover's Algorithm analysis (1996)
          └── Source: Peer-reviewed quantum computing research
              └── Multiple independent academic papers
```

Every hop provides a real, verifiable source. The chain terminates at trusted, independent origins.

Hallucination provenance chain:

```
Claim: "SG/Send processes 10,000 transfers per day"
  └── Source: LLM output (Claude session, 18 Feb 2026)
      └── Source: [DEAD END — no data supports this]
```

The chain terminates at an LLM output with no supporting evidence. The LLM generated a plausible-sounding number that has no basis in reality.

Partial provenance chain:

```
Claim: "Most enterprises use email for sensitive file transfer"
  └── Source: Industry experience (human opinion)
      └── Source: [No formal survey or study cited]
```

Not fabricated — it's probably true based on experience — but not formally grounded either. It sits in the middle of the spectrum: an informed opinion, not a verified fact.

Issues-FS Advantage

Issues-FS already captures this structure naturally. Every piece of information exists as a node. Every node can have edges to its sources. The fractal nature means we can zoom in on any claim and see its full provenance tree. The question isn't "can we do this?" — it's "how do we formalise the scoring and make it queryable?"

Source Credibility

Not all sources are equal. The graph should track source credibility over time.

What Makes a Source Credible?

The human's definition (and it's a good one):

A credible source is **not** necessarily one that is right all the time. A credible source is one that:

1. **Is consistent** in its assessments — doesn't flip-flop without reason
2. **Cites its own sources** — provides provenance for its claims
3. **Doesn't contradict itself** — or when it does, acknowledges the contradiction
4. **Changes opinion for good reasons** — when evidence changes, the opinion changes, and the reason for the change is documented
5. **Produces verifiable claims** — claims that can be independently checked

A source that changes its position because new evidence emerged is MORE credible, not less. A source that holds a position despite contradictory evidence is LESS credible.

Credibility Signals in the Graph

```
Source: "AppSec Agent"
```

```
├── Total claims: 147
├── Claims with provenance: 132 (89.8%)
├── Claims independently verified: 98 (66.7%)
├── Claims contradicted by other sources: 3 (2.0%)
├── Opinion changes: 5
│   └── All 5 backed by new evidence ✓
├── Self-contradictions: 1
│   └── Acknowledged and resolved ✓
└── Credibility score: HIGH
```

vs.

```
Source: "Random LLM output without review"
├── Total claims: 84
├── Claims with provenance: 12 (14.3%)
├── Claims independently verified: 8 (9.5%)
├── Claims contradicted by other sources: 11 (13.1%)
├── Opinion changes: 7
│   └── 2 backed by evidence, 5 unexplained
├── Self-contradictions: 6
│   └── None acknowledged
└── Credibility score: LOW
```

Over time, the graph builds a credibility profile for every source — human, agent, system, document. This is emergent, not manually assigned. The graph computes credibility from the structural properties of what the source has produced.

Hallucination and the Wardley Evolution Scale

The human makes a crucial observation: **the tolerance for hallucination should vary with the maturity level of the work.**

Evolution Stage	Hallucination Tolerance	Why
Genesis	High	We WANT creativity, speculation, "what if?" thinking. Hallucination-as-feature. Explore the space. Generate novel ideas. Some will be wrong — that's the point.
Custom Build	Medium	We're converging on something. Facts should be checked. Opinions should be labelled. But there's still room for informed speculation.
Product	Low	This is going to users. Claims must be evidence-backed. Opinions must be clearly flagged. Speculation must be removed or quarantined.
Commodity	Zero	This is infrastructure. It must be reliable. No variation, no ambiguity, no hallucination. Code, not LLM outputs. Deterministic, not probabilistic.

The Implication for LLM Usage

As a system matures along the evolution scale, the role of the LLM should diminish:

- **Genesis**: LLM generates ideas, explores possibilities, drafts initial concepts. High temperature. High creativity. Hallucination is acceptable — it's brainstorming.
- **Custom Build**: LLM assists, but outputs are verified. Evidence is required. The human (or another agent) validates.
- **Product**: LLM usage is limited to well-defined tasks with clear verification. Outputs are tested against known ground truth.
- **Commodity**: LLM is replaced by code. The system is deterministic. The simplest possible code that works 99.999% of the time. No probabilistic components.

The danger the human identifies: putting Genesis-quality systems (LLMs with no grounding) into Commodity-level production. That's where hallucination becomes harmful — not in the brainstorming phase, but in the production phase where reliability is expected.

How Issues-FS Captures This

Node-Level Metadata

Every node (fact, claim, opinion, output) carries:

```yaml
content: "AES-256-GCM provides 128-bit security against quantum attacks"
source: "architect-agent-session-2026-02-19"
source_type: "llm_agent"
provenance:
  - type: "reference"
    target: "nist-pqc-faq-2024"
    verified: true
  - type: "reference"
    target: "grovers-algorithm-analysis"
    verified: true
grounding_score: 0.92
contradiction_count: 0
evolution_stage: "product"
hallucination_flag: false
confidence: "high"
last_verified: "2026-02-19"
```

Edge-Level Metadata

The relationships between nodes also carry hallucination-relevant data:

```yaml
```

```yaml
edge:
  from: "claim-aes-256-quantum"
  to: "nist-pqc-faq-2024"
  relationship: "supported_by"
  strength: "strong"    # strong / moderate / weak / contradicted
  verified_by: "human-review-2026-02-19"
```

Queryable Hallucination Graph

With this metadata, we can query:

- **"Show me all ungrounded claims"** — nodes with grounding_score < threshold, no provenance edges
- **"Show me all contradictions"** — pairs of nodes that contradict each other
- **"Show me the credibility of source X"** — aggregate credibility score across all claims from that source
- **"Show me hallucination risk by evolution stage"** — are there ungrounded claims in Product or Commodity-stage work?
- **"Show me the provenance tree for claim X"** — trace back through every supporting source
- **"Show me everything that depends on an ungrounded claim"** — if a hallucinated fact is the foundation, what's built on top of it?

That last query is particularly powerful: it reveals the blast radius of a hallucination. If a fabricated number is cited as the basis for a budget decision, which in turn drives a resource allocation, which in turn affects a product feature — the graph shows the entire chain of dependence.

LLM Comparison as Experimental Data

The human suggests using LLM outputs as experimental data for hallucination detection. This is a strong idea:

- Give the same prompt to multiple LLMs (or the same LLM at different temperatures)
- Capture all outputs as nodes in the graph
- Compare: where do they agree? (Higher confidence — multiple independent sources converge.) Where do they disagree? (Lower confidence — investigate further.) Where does one produce a claim with no equivalent in others? (Potential hallucination.)
- Vary the temperature: at low temperature, outputs should be more grounded. At high temperature, more creative but less reliable. The graph should reflect this — the grounding scores should correlate with temperature.

This is a research experiment: **can the graph structurally detect hallucination by comparing multiple LLM outputs?** The hypothesis is yes — hallucinated content will show up as orphan nodes with no cross-model agreement and no external provenance.

People Hallucinate Too

A final observation from the human: hallucination is not an LLM-specific problem. People hallucinate — they state unfounded opinions as facts, misremember, make things up (sometimes unintentionally), and present speculation as certainty.

The graph should treat all sources equally. A human claim without evidence is no more grounded than an LLM claim without evidence. The provenance chain doesn't care about the source type — it cares about the evidence.

Companies hallucinate too — marketing materials, earnings projections, press releases. Entire organisations can operate on ungrounded assumptions that nobody has traced back to evidence.

The Issues-FS hallucination detection model applies to all of these: trace the provenance, measure the grounding, score the credibility, flag the dead ends.

Implementation Roadmap

Task	Priority	Maturity
Define the grounding score formula (inputs: provenance depth, source count, source credibility, contradictions)	P2	Genesis
Add grounding metadata to the node schema (provenance edges, source type, confidence, hallucination flag)	P2	Custom Build
Implement provenance chain traversal (trace any claim back to its sources)	P2	Custom Build
Implement contradiction detection (find pairs of nodes that conflict)	P3	Genesis
Implement source credibility scoring (aggregate from historical claims)	P3	Genesis
Build the "hallucination risk by evolution stage" query	P3	Custom Build
Build the "blast radius" query (what depends on an ungrounded claim?)	P3	Custom Build
Design the LLM comparison experiment (multi-model, multi-temperature)	P4	Genesis
Visualisation: hallucination heat map across the graph	P4	Genesis

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The Big Idea

Every piece of information in the system exists on a spectrum from fully grounded to completely fabricated. The graph doesn't just store information — it stores the evidence for that information, the credibility of the sources, and the confidence we should have in each claim.

When you query the graph, you don't just get answers. You get answers with confidence scores, provenance chains, and contradiction flags. You can ask: "How sure are we about this?" and the graph can tell you — structurally, not rhetorically.

This is what "just words" becomes when you add connectivity: meaning, evidence, confidence, and the ability to distinguish what we know from what we think from what we made up.
